Rodolfo's Encyclopedia

Rodolfo e Seus Capangas

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Part I

Number Theory

Chapter 1

Basic Theory

1.1 Divisibility

Theorem 1. *If $x, y \in \mathbb{Z}$, and $x \mid y$ then $y = 0$ or $\mid x \mid \leq \mid y \mid$*

Theorem 2. *Suppose $c, x, y \in \mathbb{Z}$, and $c \mid x, y$ then $c \mid ax + by$ for any $a, b \in \mathbb{Z}$*

Definition 3.

Theorem 4. *Let a, b, c be three integers. Then*

$$c \mid a, c \mid b, \Rightarrow c \mid \gcd(a, b)$$

Theorem 5.

Theorem 6.

1.2 Base Systems

Definition 7. Given a positive integer b, the base$-b$ representation of a real number r is a string of digits:

$$(a_n a_{n-1} \dots a_1 a_0 . a_{-1} a_{-2} \dots)_b$$

Where $a_i \in \{0, 1, \dots, b-1\}$, and

$$r = a_n b^n + a_{n-1} b^{n-1} \dots a_1 b^1 + a_0 b^0 + a_{-1} b^{-1} a_{-2} b^{-2} \dots$$

Theorem 8. *The base$-b$ representation of any real number is unique.*

1.3 Bézouts' Theorem

Chapter 2

Basic Modular Arithmetic

2.1 Properties

2.2 Fermat's Little Theorem

Theorem 9.

2.3 Euler's Theorem

Chapter 3

Arithmetic Funtions

3.1 Number of Divisors

3.2 Sum of Divisors

3.3 Arithmetic Funtions

3.4 Multiplicative Functions

3.4.1 Möbius Inversion

3.4.2 Dirichlet Convolution

3.5 Floor and Ceiling Functions

Chapter 4

Binomial Coefficients

Definition 10. For non-negative integers a, b, we define

$$\binom{a}{b} = \begin{cases} 0 \text{ if } a < b \\ \dfrac{a!}{b!(a-b)!} \end{cases}$$

Theorem 11 (Lucas Theorem)**.**

Chapter 5

Diophantine Equations

5.1 Pythagorean triples

5.2 Fermat's Descent

5.3 Pell's equation

5.4 Vieta's Jump

Chapter 6

Polynomials

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6.2 GCD

6.3 Vietas Theorem

6.4 Irreducibility

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Advanced Modular Arithmetic

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Chapter 8

Prime Exponents

8.1 p-adic valuations

8.2 LTE

8.3 Zsigmondy's Theorem

Chapter 9

Quadratic Residues

9.1 Quadratic Reciprocity